

THEMIS Deployment Manual

0. Introduction

0.1 Current Version

- Manual version: v0.3.2
- Compatible Minimum AOS version: v0.3.2
- Compatible Minimum AOS Server version: v0.3.2

0.2 About This Document

This document is based on the THEMIS humanoid robot system running AOS, and describes how to set up and run the demonstration workflow. The demonstration works as follows: the robot picks up a target object at point A (this case uses a can of Coke or Sprite as an example), carries the object to point B and places it down (a blue tote box is used as an example), then goes to point C to greet the audience, and finally returns to point A — repeating the cycle continuously.

1. Preparation

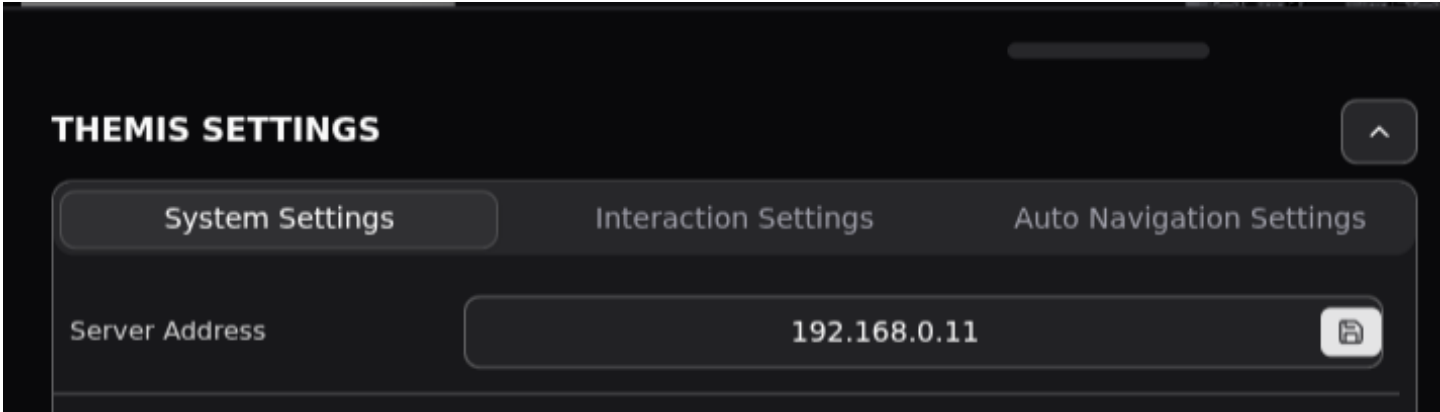
- Make sure the control device, such as a tablet or computer, and the robot are connected to the same LAN.
- Open the browser on the device and enter the IP address followed by port 5000. Using the default IP address 192.168.0.11 as an example, the address to enter in the browser is:
<http://192.168.0.11:5000/>
- After entering the address above, you will land on the initialization page by default.
- Print the AprilTag navigation markers from: [AprilTag_0-20_family36h11.pdf](#). Note: do not use the marker with ID 0.

2. Initialization

2.1 Before use, please confirm the following:

- A data connection has been established between this device and the robot (the black main button on the left is clickable; if it is greyed out, first go to **Settings** to check the network and the server, and only proceed once the connection is successful).

- Before performing the **Initialize** step, the robot must be placed safely and stably on the ground (A confirmation prompt will appear before this step is executed.).



- Click the gear icon on the top-right corner, open the settings drawer from the bottom of the screen and enter the robot's IP address (the default IP is 192.168.0.11) in the Server Address field.

2.2 Main Functions:

 同步块 [该内容不支持导出查看]

Initialization Page Overview

- The initialization page is mainly used to start the robot's subsystems in a fixed order (Boot up -> Run BEAR -> Initialize -> Estimation -> Control -> Locomotion).
- The main button on the left executes the current step, and the step list on the right shows the progress. The 3D robot model in the background updates in real time according to the joint and body states.
- Once all steps have completed successfully, the system automatically enters the main control interface.
- Other functions:

Module Location	Function	When to Use
Right background — Robot model	A 3D preview of the robot's posture during the startup phase, updated in real time with joint/body data	Only for observing whether the posture is normal; it cannot replace the main action button on the left for completing the startup steps
Top-right — Red warning icon	Appears only when there is a BEAR driver fault or an abnormal thread state; click to view the error type and the motor IDs	The icon will keep flashing when there are motor errors, motor overheating, thread errors, startup-step errors, etc.;

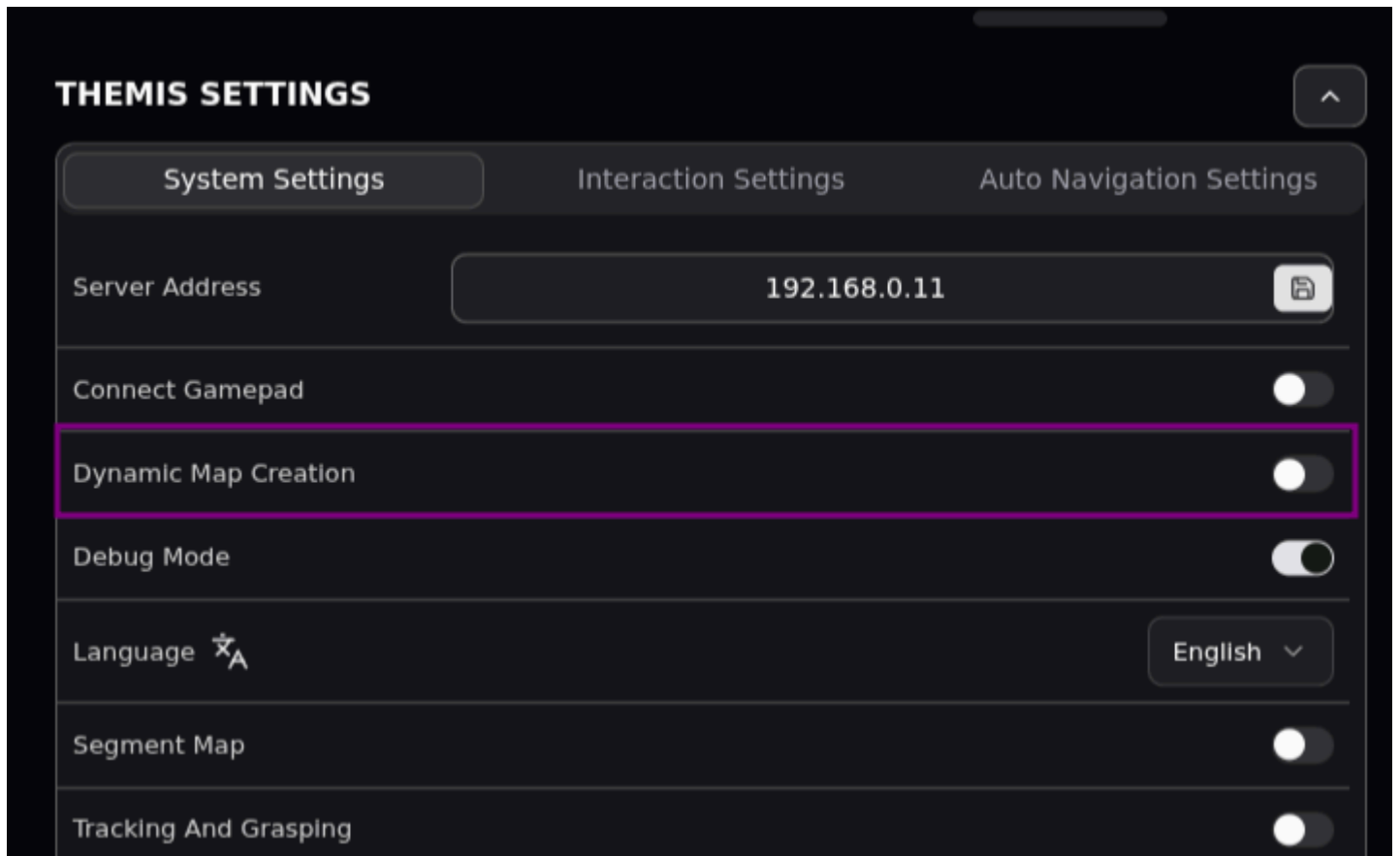
	involved. For some absolute-position errors, you can attempt to resolve the issue directly from the pop-up window.	no icon means none of the above errors are present
Top-right — Wrench icon	Maintenance functions such as motor calibration, IMU calibration, and joint-limit inspection	Not part of the standard startup procedure. Use as needed when you find something that requires calibration or configuration
Top-right — Gear icon	Opens the THEMIS settings drawer menu from the bottom, which includes system parameters, process monitoring (Watchdog), restarting the data/video services, shutdown, etc.	Use when you cannot connect, need to change the server IP/language, or need to restart a service
Top-right — Skip icon	Goes straight to the main control interface without executing the remaining robot startup steps	Use this when you need to skip the remaining startup steps

3. Mapping and Waypoint Confirmation

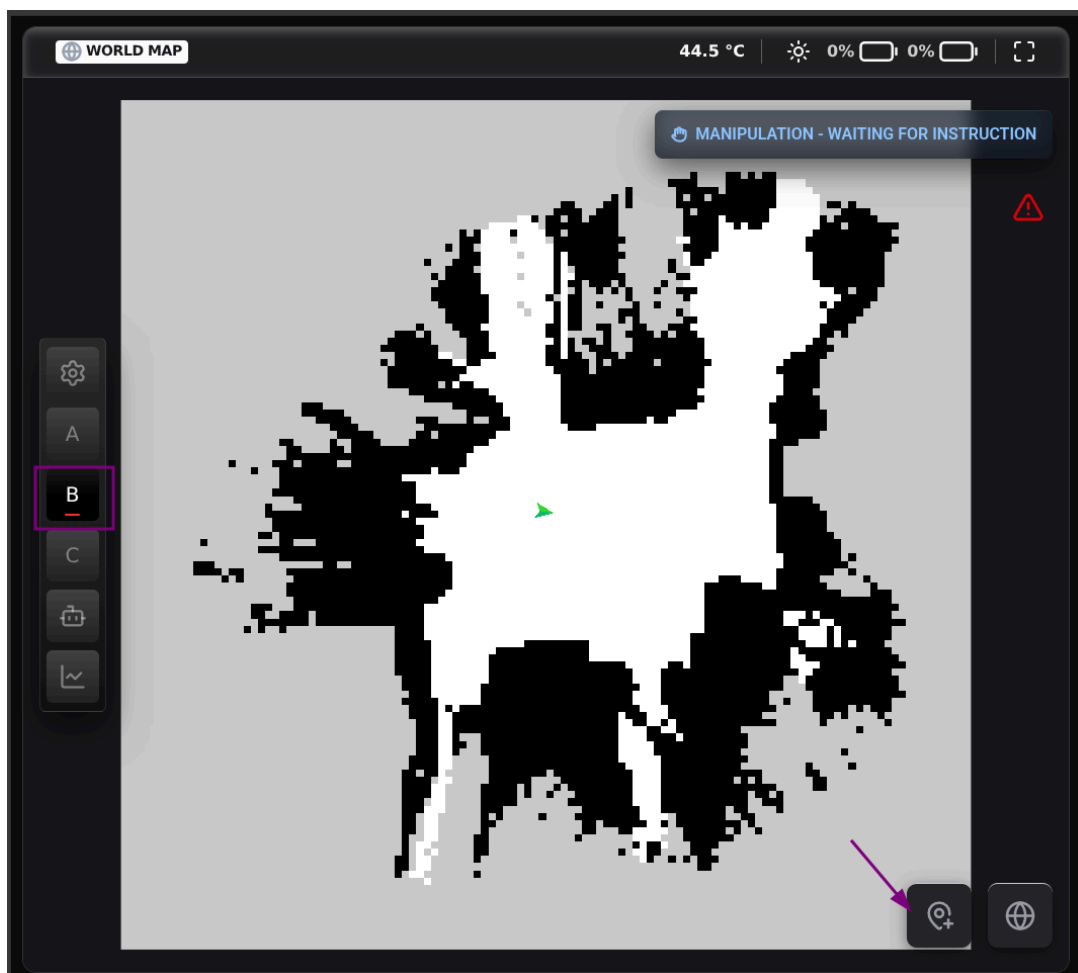
Note: When placing the AprilTag markers during mapping, make sure that the grasp point, the placement point, and the greeting point each have a corresponding AprilTag markers. Note: do not use the AprilTag markers with ID 0.

3.1 Building the Global Map

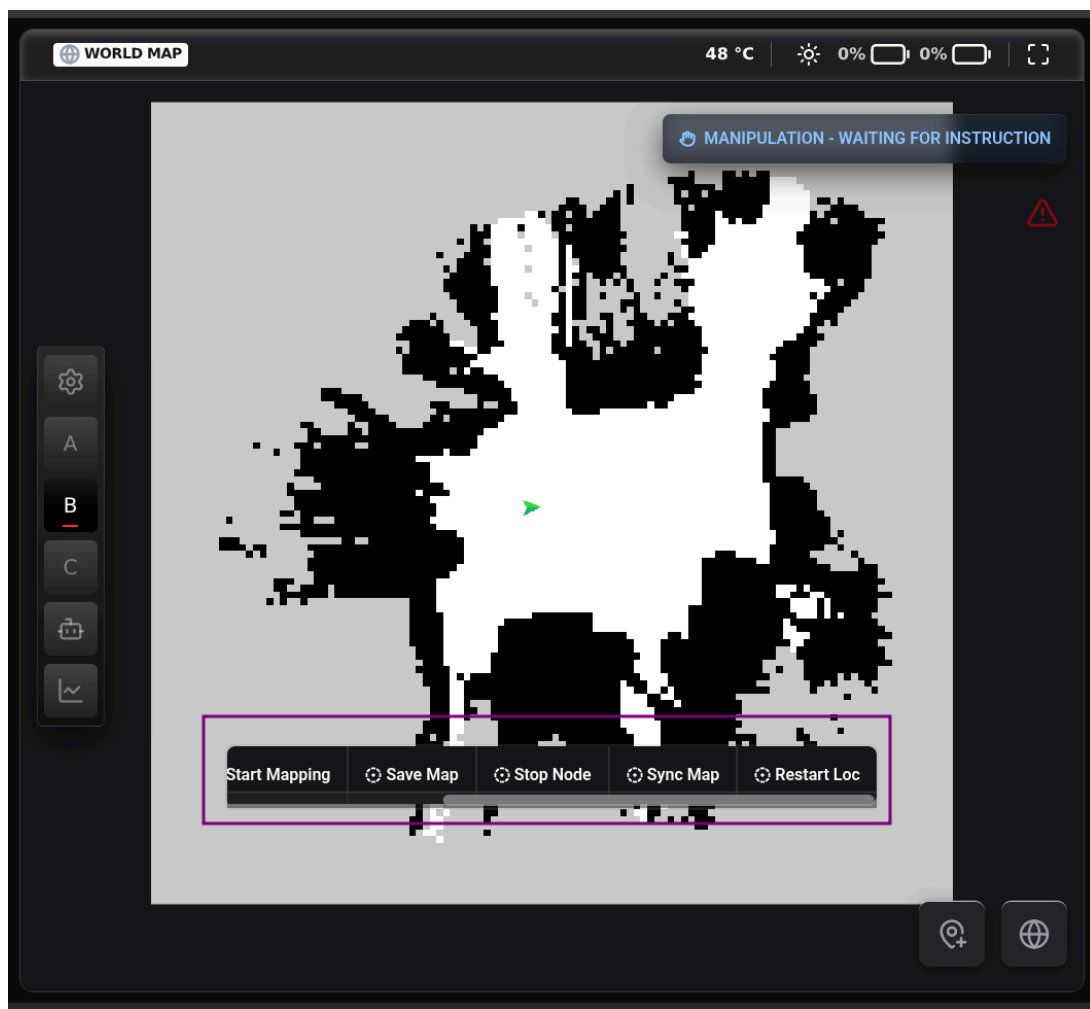
1. Manually position and orient the robot so that it faces the grasp point (the placement point also works — this makes it easier to calculate the yaw angle of the three points later). Then open **System Settings** and turn on the **Dynamic Map Creation** toggle.



2. Go to Interface B and click the button in the bottom-right corner; the dynamic map creation step menu will appear. Then follow the steps to build the map. When mapping is complete, click Save Map, Stop Node, and Sync Map in turn.



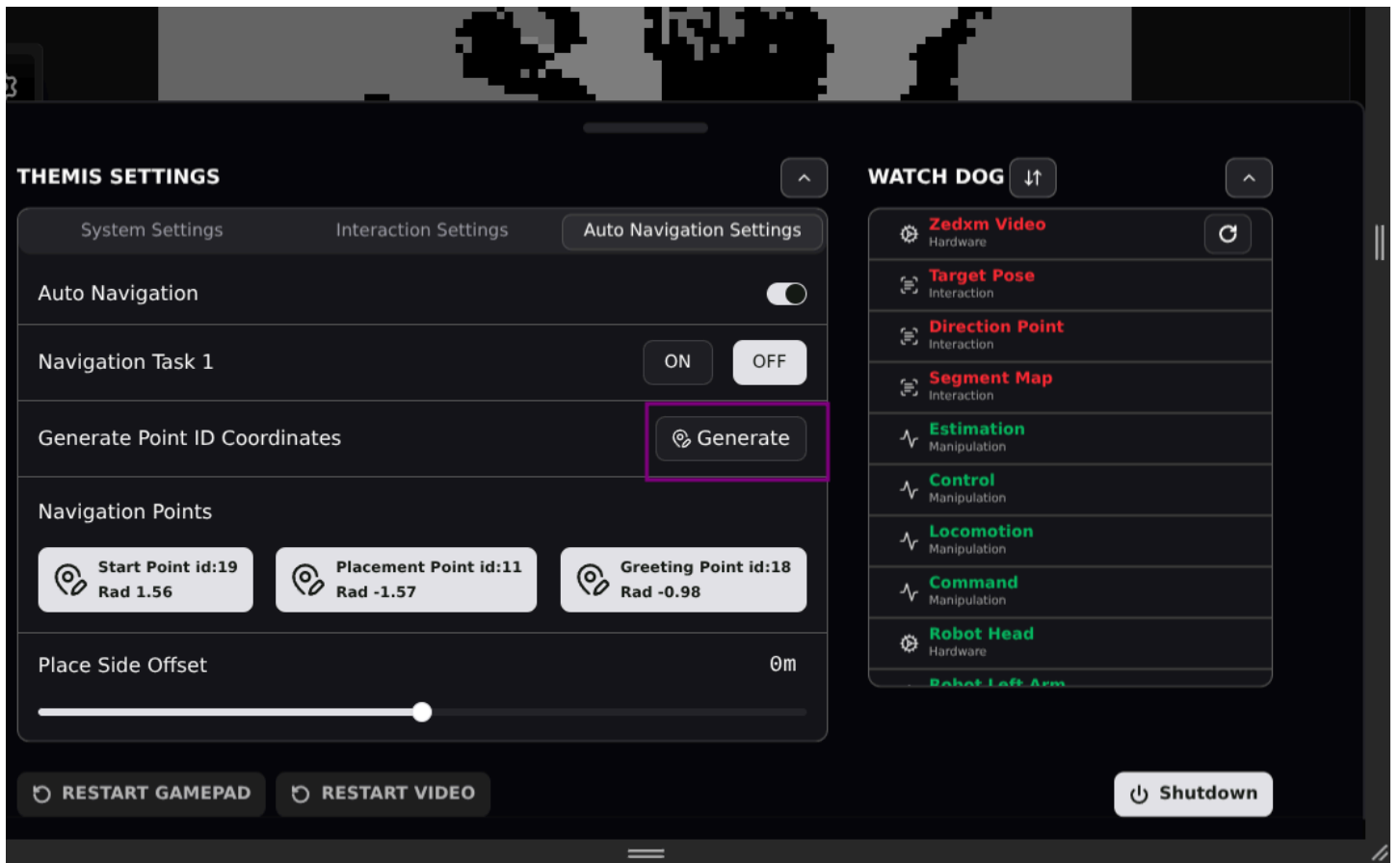
Dynamic Map Creation Menu



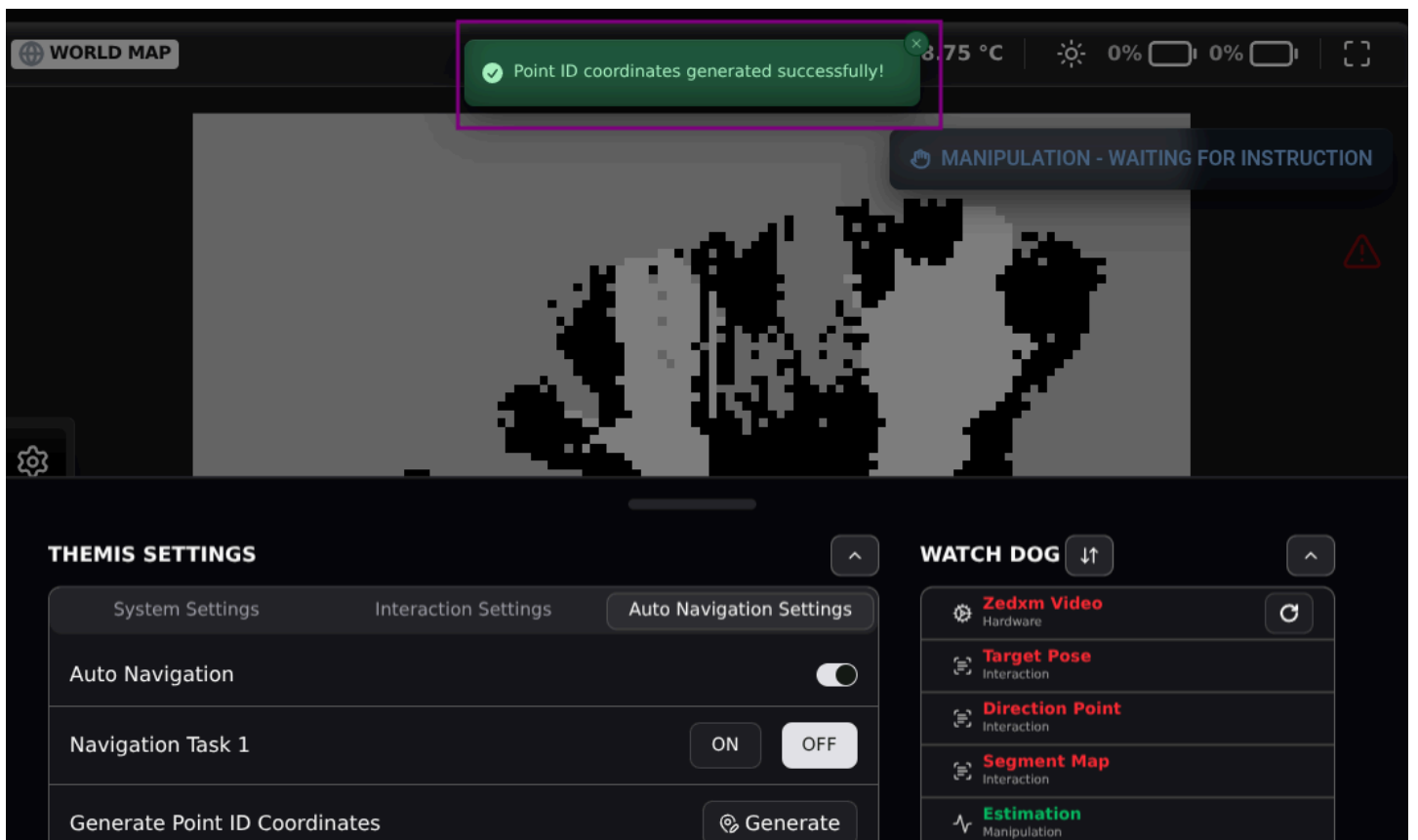
Dynamic Map Creation Process

3.2 Waypoint Setup

1. After creating a new map, you need to select the waypoints. Before doing so, the waypoints must be generated manually (after each mapping session you only need to generate them once, and all ID coordinates will be saved). Click the **Generate** button in Auto Navigation Settings to generate the waypoint ID coordinates.

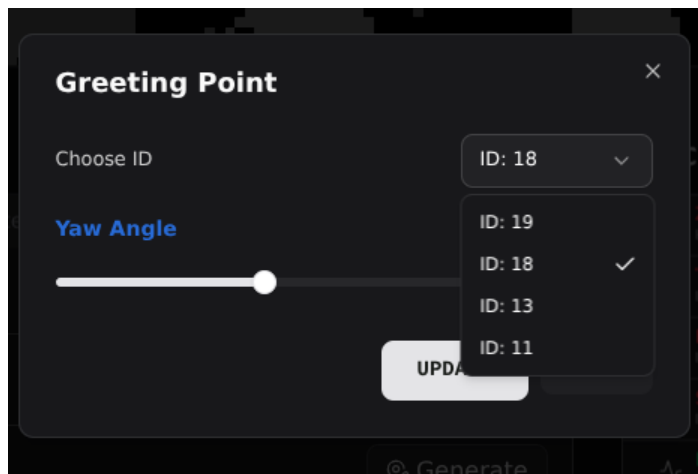


Click Generate under Generate Point ID Coordinates.

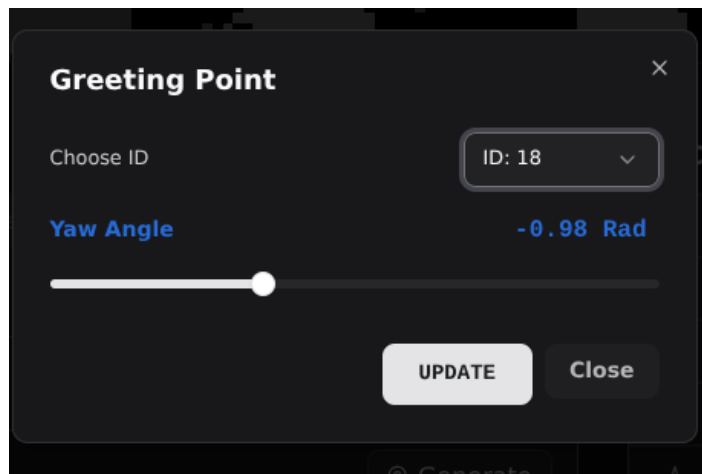


Successful Generation Confirmation

2. Select the waypoints: click the greeting point, then choose the **Yaw Angle** for each point. The heading at the position where mapping began is taken as 0 rad, and the yaw angle for each position is then chosen according to the right-hand rule. (For example, if the robot was facing the grasp point when mapping started, then the yaw of the grasp point is 0.)



Select the ID of the Greeting Point



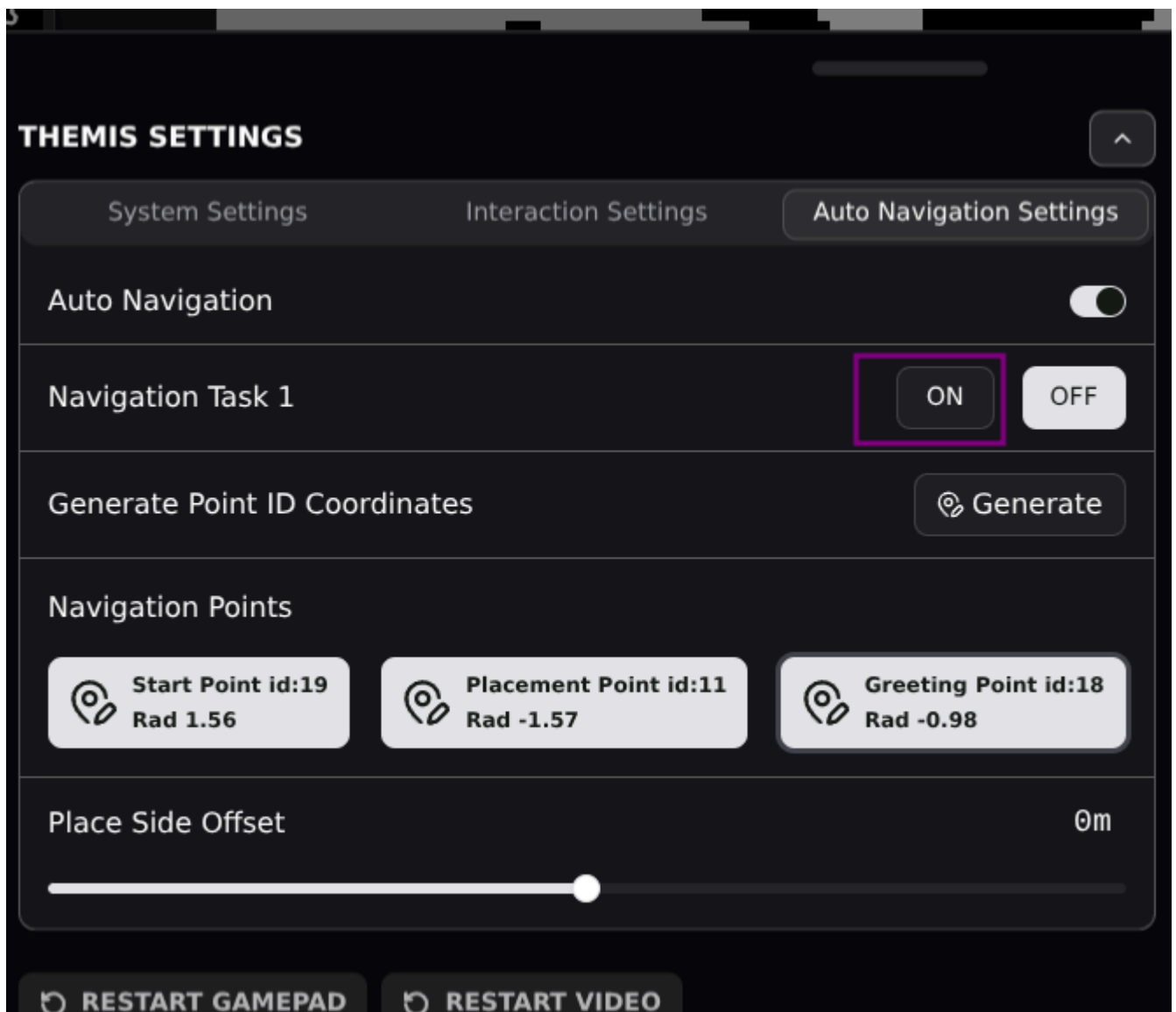
Modify the Yaw Angle of the Greeting Point

4. Starting and Stopping the Auto Navigation Workflow

4.1 The steps for Starting and Stopping the Workflow of Auto Navigation Task

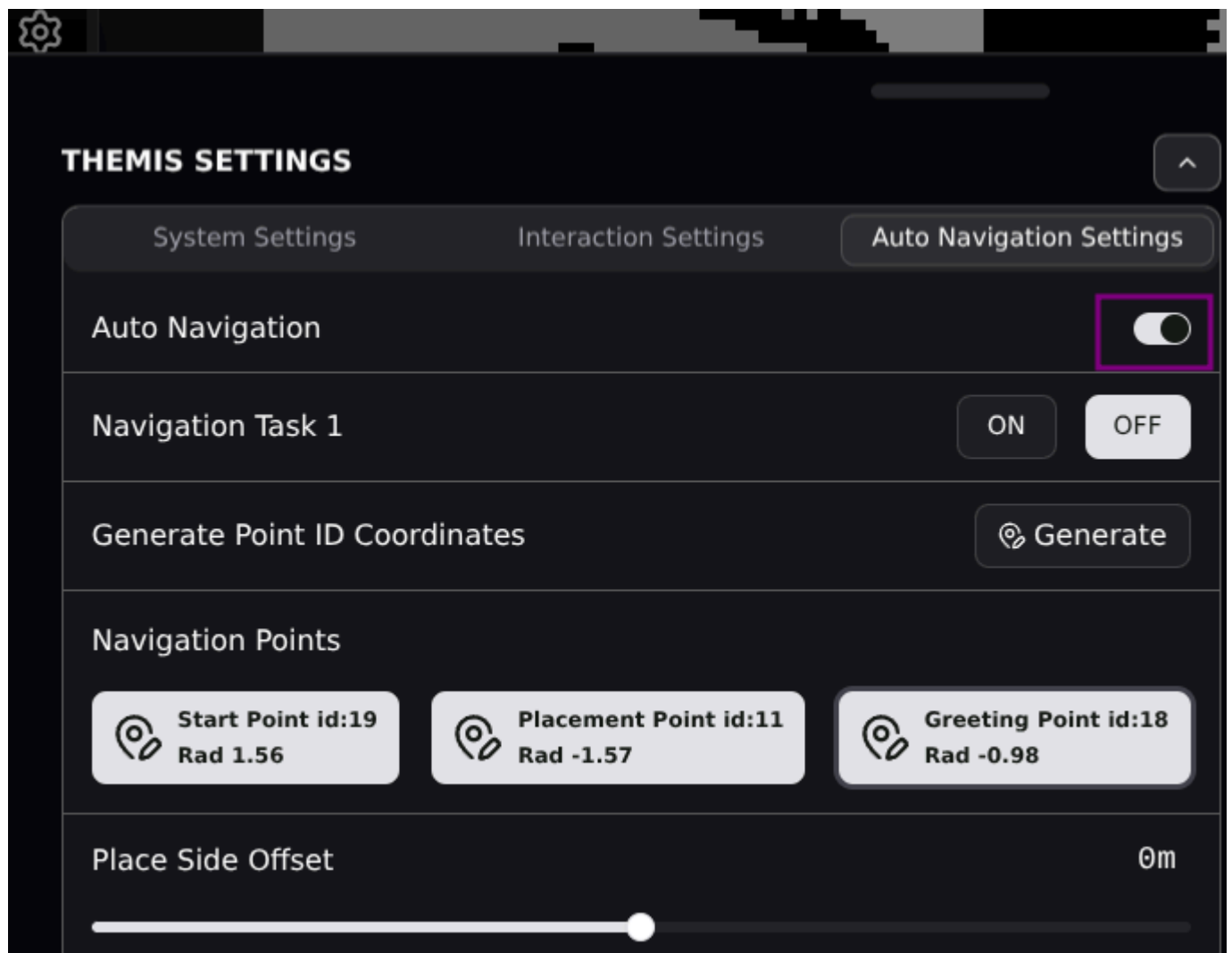
1. Click the gear icon on the left and select Auto Navigation Settings.

Click ON next to Navigation Task 1. At this point, the global map should be visible in **Interface B**, and a path to the grasp point should be planned on the global map.



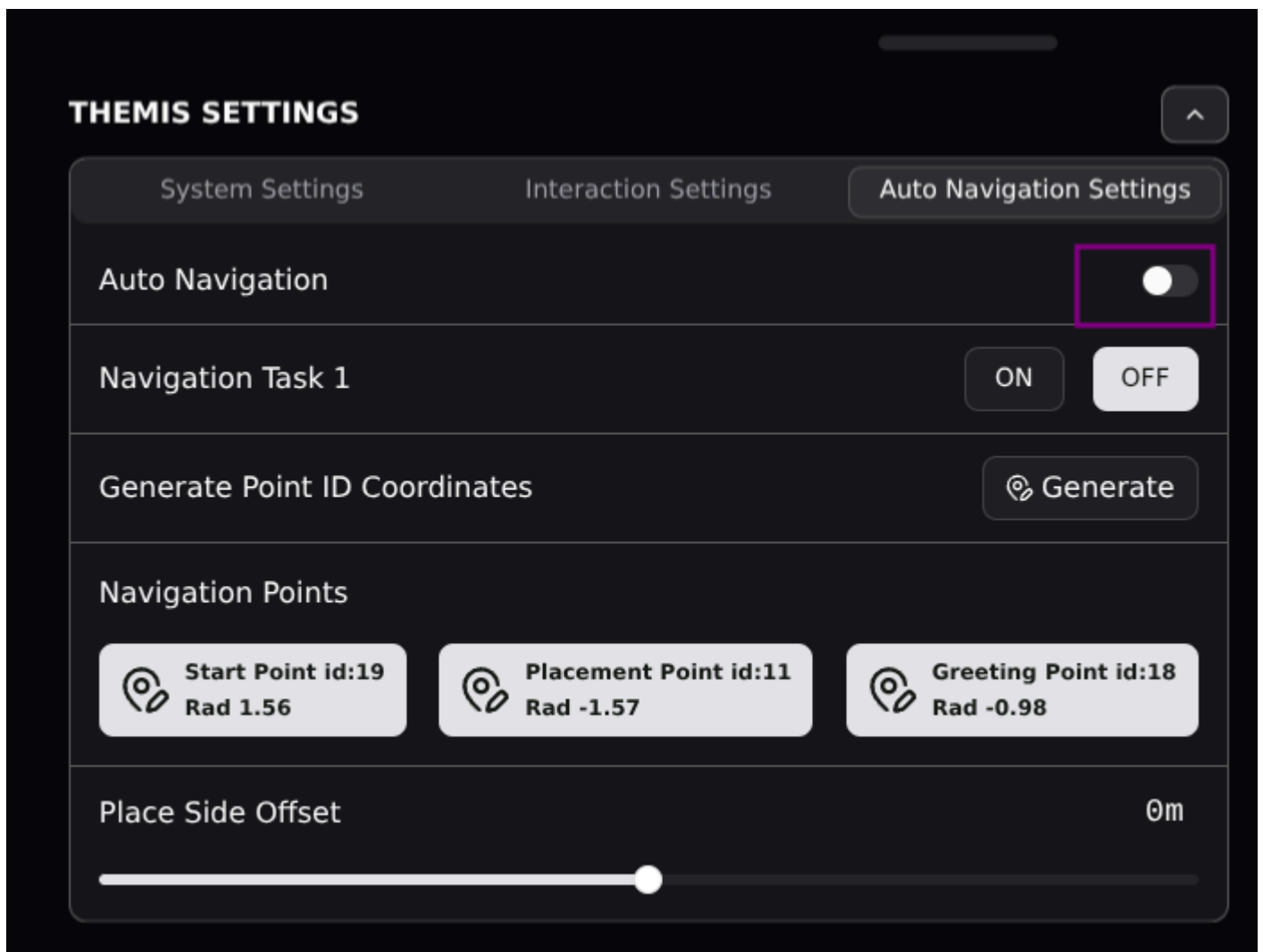
Turn on the Navigation Task

2. Once the path appears, the robot needs to be switched to auto navigation mode before it can start performing the task. Enable it in Auto Navigation Settings.



Turn on the Auto Navigation Mode

3. To exit Auto Navigation, turn the Auto Navigation toggle off. The robot will begin stepping in place. Press **Start + Back** on the gamepad to stop the robot and return it to General Mode.



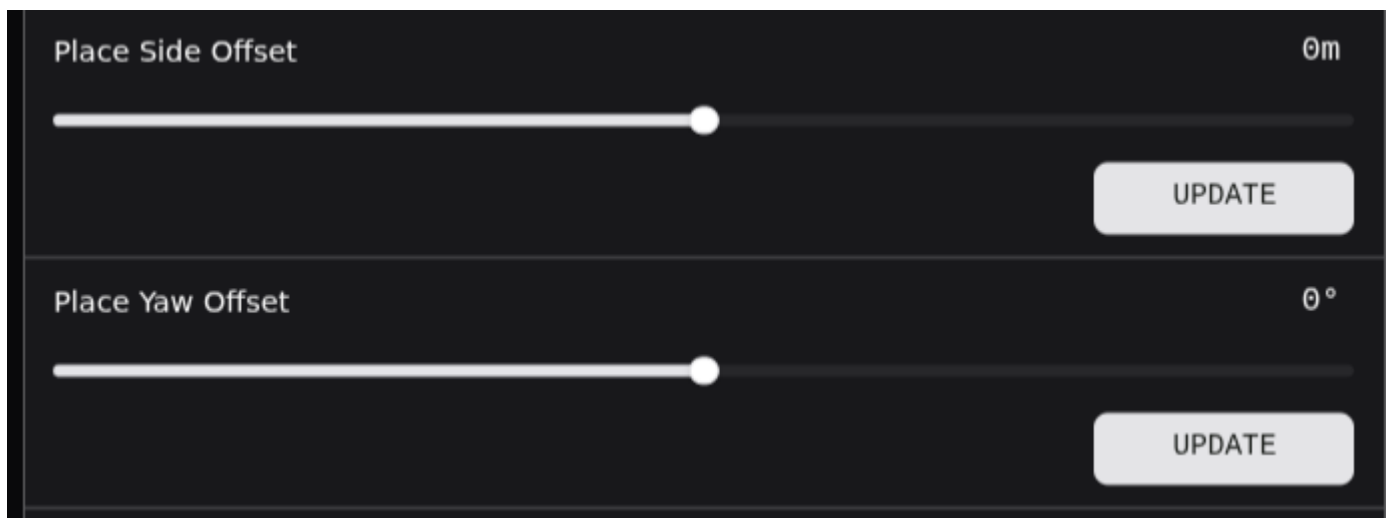
Turn off the Auto Navigation Mode



If the robot moves toward the grasp point but does not lower its head to look at the navigation waypoint after starting Navigation Task 1, click ON for Navigation Task 1 again, then toggle Auto Navigation off and back on. Confirm that the robot lowers its head toward the navigation marker.

4.2 Parameter Settings

1. Place Side Offset: The range is -0.5 m to 0.5 m. If set to 0, there is no offset when the robot arrives at the placement point. If set to a positive value, e.g. 0.1 m, the target navigation point is offset according to which hand the robot uses to grasp the object: if the object is grasped with the left hand, the navigation position is shifted to the right (relative to the navigation point's coordinate frame); if grasped with the right hand, the navigation position is shifted to the left. A negative value produces the opposite effect.
2. Place Yaw Offset: If a positive angle is used, e.g. 60°, then when the robot navigates to the placement point it subtracts the current Yaw offset from the navigation point's Yaw value, so that the robot stands sideways relative to the container.

The image shows a dark-themed user interface for 'Placement Parameter Settings'. It contains two sections. The first section, 'Place Side Offset', has a horizontal slider with a white knob positioned at approximately the 50% mark. To the right of the slider is a value display showing '0m' and an 'UPDATE' button. The second section, 'Place Yaw Offset', has a similar horizontal slider with its knob also at the 50% mark. To its right is a value display showing '0°' and another 'UPDATE' button.

Place Side Offset 0m

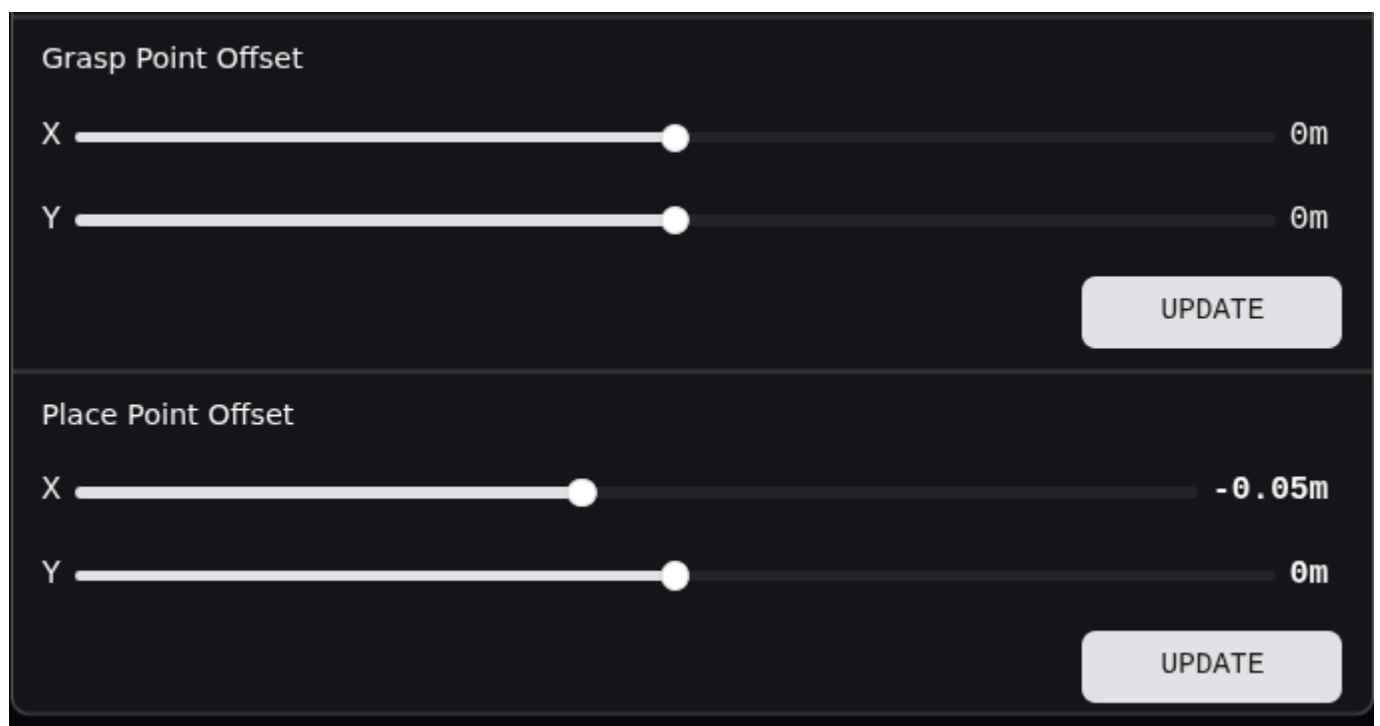
UPDATE

Place Yaw Offset 0°

UPDATE

Placement Parameter Settings

3. Grasp-point and placement-point offsets: This offset is based on the local coordinate frame of the corresponding grasp or placement waypoint. If X is negative, the target position is shifted backward relative to the corresponding grasp or placement waypoint; if Y is set to a negative value, the navigation position is shifted to the right by the distance you set.

The image shows a dark-themed user interface for 'Grasp-Point and Placement-Point Offset Settings'. It is divided into two main sections. The top section, 'Grasp Point Offset', contains two horizontal sliders. The 'X' slider has its knob at the 50% mark with a value display of '0m'. The 'Y' slider also has its knob at the 50% mark with a value display of '0m'. An 'UPDATE' button is located to the right of these sliders. The bottom section, 'Place Point Offset', also contains two horizontal sliders. The 'X' slider has its knob shifted to the left, with a value display of '-0.05m'. The 'Y' slider has its knob at the 50% mark with a value display of '0m'. An 'UPDATE' button is located to the right of these sliders.

Grasp Point Offset

X 0m

Y 0m

UPDATE

Place Point Offset

X -0.05m

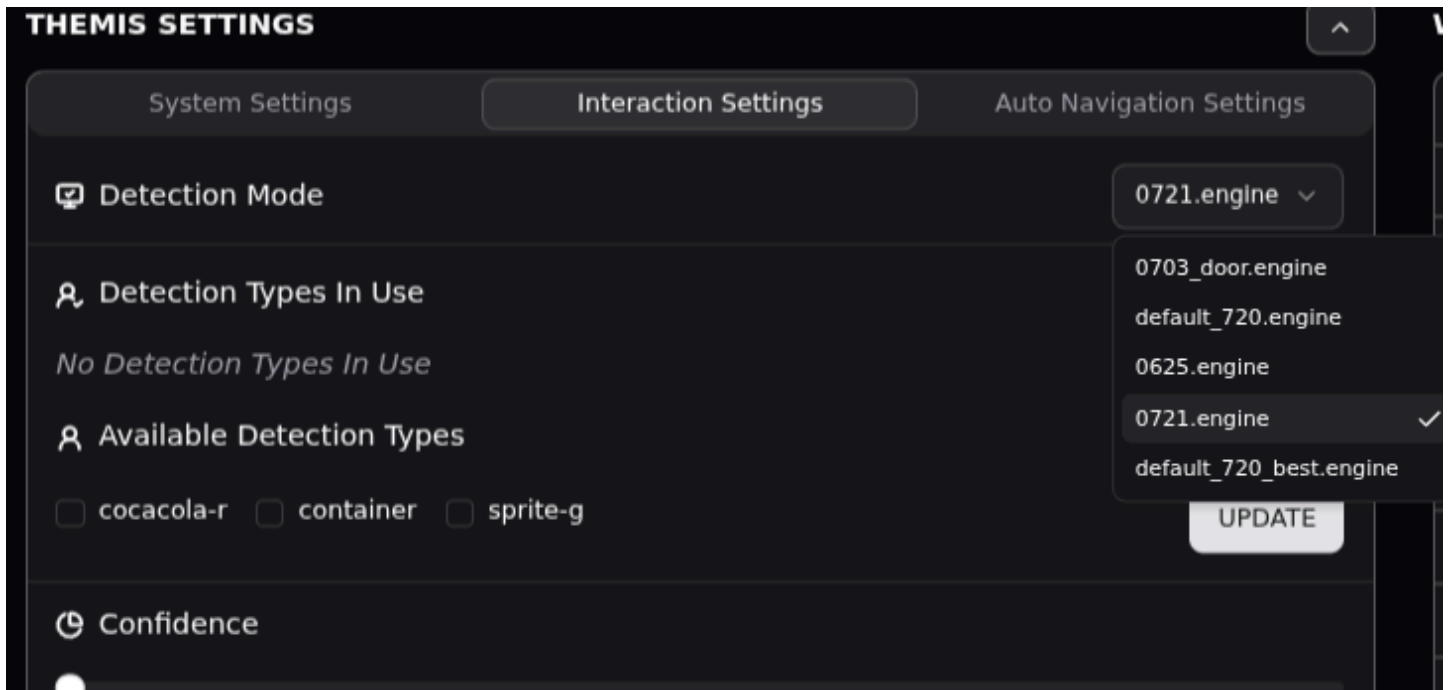
Y 0m

UPDATE

Grasp-Point and Placement-Point Offset Settings

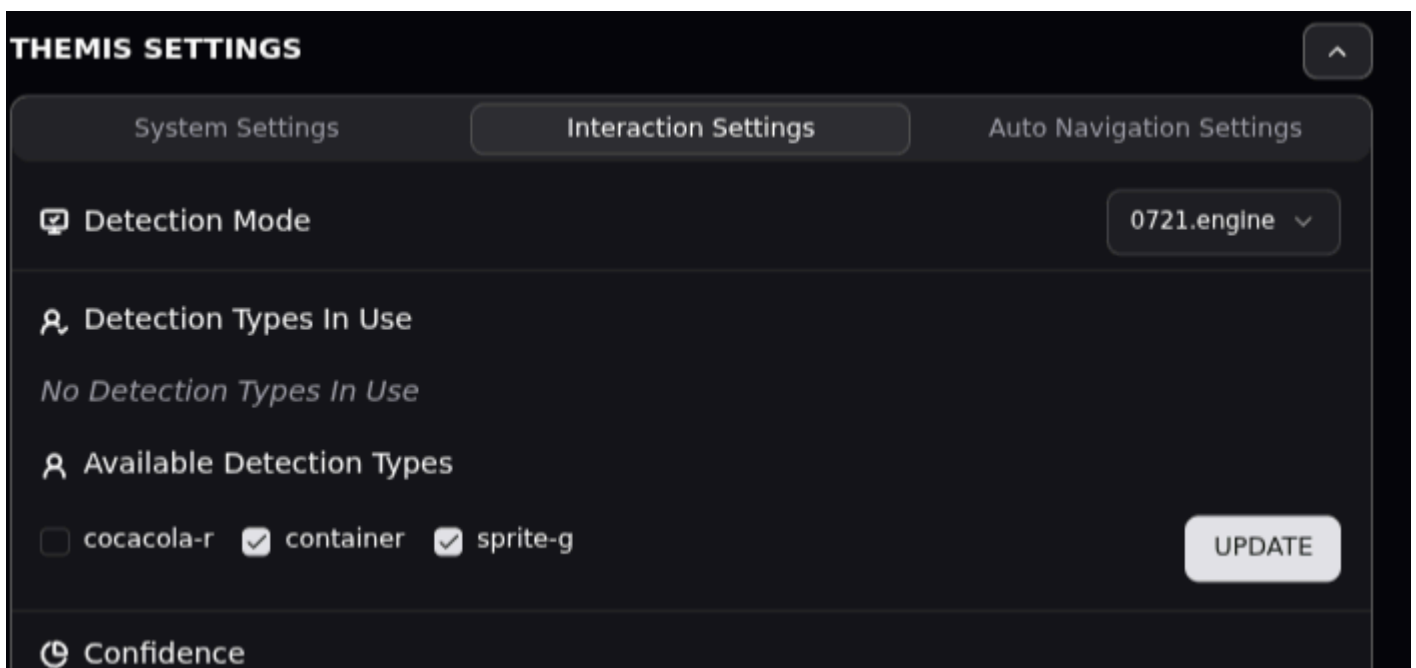
5. Detection Mode and Type Selection

The detection mode and detection types can be configured in the **Interaction Settings** panel. To switch the detection mode, use the drop-down menu in the top-right corner of the figure below to select an existing mode weight file (switching models takes roughly 20–40 s).



Switch the Detection Model

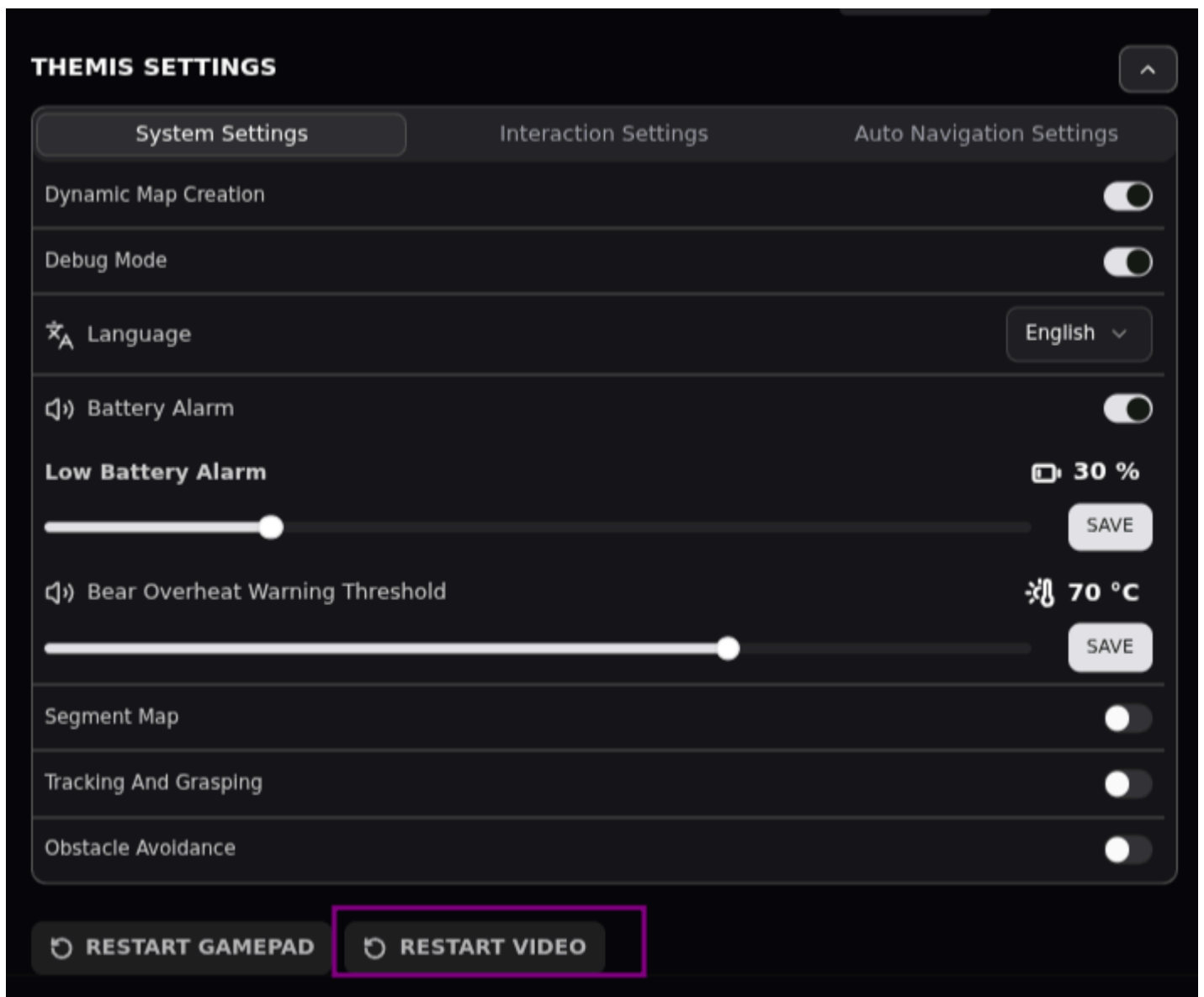
In the **Available Detection Types** list, tick the detection types you want and click Update Settings. The system will detect only the selected object types. (If all are deselected, it detects all categories by default.)



6. Other Issues

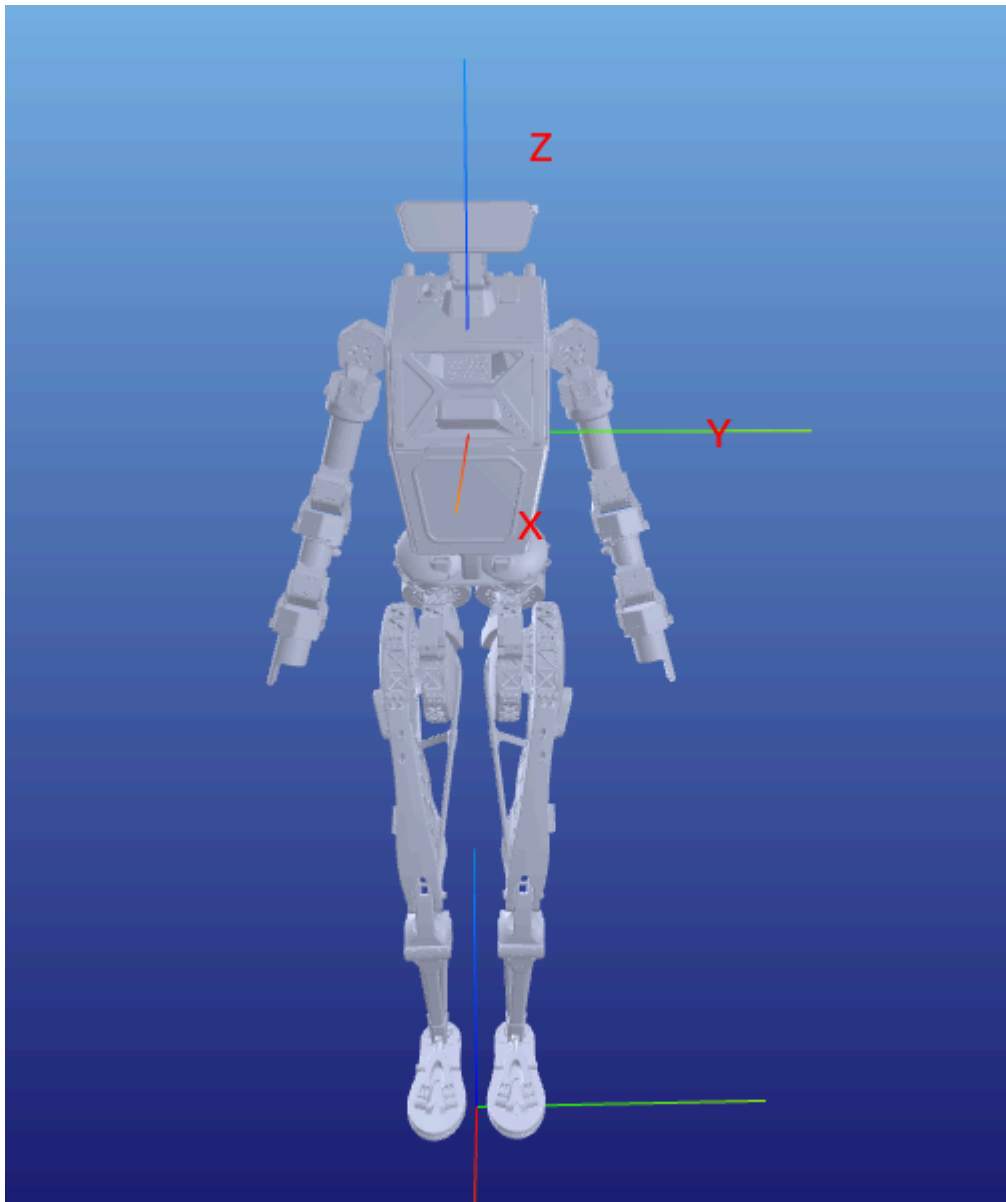
- Video Stream Issues

For a short period after the robot is powered on, the video stream may be unavailable. This is because the data inside the robot is temporarily out of sync. In this case, simply click the **Restart Video** button, located at the very bottom of the pop-up menu, then wait about 20 seconds and refresh the browser.



Restarting the Video Stream Service

- Grasping Issues

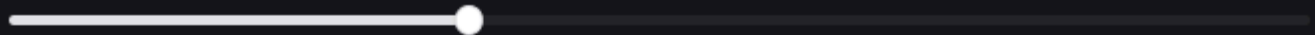


1. The grasp point is expressed relative to THEMIS's BASE_LINK. The definition of the BASE_LINK coordinate frame (right-hand rule) is shown in the figure above.
 - During grasping, if you find that the palm is offset in the positive Y direction relative to the object being grasped, you can correct this by setting the grasp parameter Y in the figure below to -m (the value should be adjusted according to the actual offset). Apply the same correction principle to the X- and Z-axis offsets.
2. Recommended grasp-speed setting: 50%–70%.



Manipulation Velocity

35 %



UPDATE



Manipulation Offsets

X  0 cm

Y  0 cm

Z  0 cm

UPDATE